First-Order Logic (FOL) — 15 Master's MCQs

COM759 Knowledge Engineering & AI | Grounded Assessment

Question 1: How is a predicate formally distinguished from a proposition in First-Order Logic (FOL)?

A) A predicate is an atomic statement without variables that evaluates directly to True or False.

B) A predicate is a verb phrase template with argument variables that maps individuals from a universe of discourse to truth values {True, False}.

C) A predicate is a zero-arity functional mapping that always returns a ground term.

D) A predicate is a sub-symbolic weight vector operating in continuous vector space.

Correct Answer: B) A predicate is a verb phrase template with argument variables that maps individuals from a universe of discourse to truth values {True, False}.

Explanation: Passages [2] & [6] define a predicate as a verb phrase template describing properties or relationships among objects. While a predicate with variables is not a proposition on its own, assigning domain values to all its variables yields a proposition.

Question 2: According to the formal syntax of FOL, which of the following expressions represents a valid ground term?

A) $P(x) \rightarrow \exists y\, Q(x, y)$

B) mark(best_friend(Bruno), MathLogic)

C) $\exists x\, (A(x) \vee B(y))$

D) Brother(John, Fred)

Correct Answer: B) mark(best_friend(Bruno), MathLogic)

Explanation: Passages [14] & [25] specify that a term denotes an object/individual. A ground term is a term containing no variables. Here, mark(best_friend(Bruno), MathLogic) applies function symbols to constants without variables, making it a ground term. Brother(John, Fred) is an atomic predicate formula, not a term.

Question 3: Given n individuals and m locations, how does predicate logic reduce spatial complexity compared to propositional logic when representing a movement template Move(x, y, z)?

A) Propositional logic requires n × m symbols; FOL requires n^2m symbols.

B) Propositional logic requires $n \cdot m^2$ distinct symbols; FOL uses a single ternary predicate template Move(x, y, z).

C) FOL requires n + m propositional constants, whereas propositional logic requires zero.

D) FOL converts $n \cdot m^2$ symbols into 3 zero-arity constants.

Correct Answer: B) Propositional logic requires $n \cdot m^2$ distinct symbols; FOL uses a single ternary predicate template Move(x, y, z).

Explanation: Passage [4] notes that representing person x moving from location y to z across n individuals and m locations requires $n \cdot m^2$ separate symbols in propositional logic, whereas FOL handles this compactly via one ternary predicate Move(x, y, z).

Question 4: What is the semantic difference between $\forall x$ (WorksAt(FBK, x) $\rightarrow$ Smart(x)) and $\forall x$ (WorksAt(FBK, x) $\wedge$ Smart(x))?

A) The former asserts everyone works at FBK and is smart; the latter asserts everyone working at FBK is smart.

B) The former asserts everyone working at FBK is smart; the latter asserts that everyone in the universe works at FBK and is smart.

C) Both formulas are logically equivalent due to De Morgan's laws.

D) The former is unsatisfiable, whereas the latter is a valid tautology.

Correct Answer: B) The former asserts everyone working at FBK is smart; the latter asserts that everyone in the universe works at FBK and is smart.

Explanation: Passages [15] & [18] explicitly caution that using $\wedge$ with $\forall$ asserts that every individual in the entire universe satisfies both conditions. Using $\rightarrow$ correctly restricts the scope of Smart(x) to individuals who work at FBK.

Question 5: Why is representing 'There is a smart student' as $\exists x$ (Student(x) $\rightarrow$ Smart(x)) considered a common representation error in FOL?

A) Because the formula evaluates to False if any student in the domain is not smart.

B) Because if there exists any individual in the universe who is NOT a student, Student(x) is False, making the implication True and vacuously satisfying the existential sentence.

C) Because existential quantifiers cannot be combined with material implications in well-formed formulas.

D) Because the formula requires x to be a ground term rather than a variable.

Correct Answer: B) Because if there exists any individual in the universe who is NOT a student, Student(x) is False, making the implication True and vacuously satisfying the existential sentence.

Explanation: Passages [15] & [18] point out that $\exists x$ (P(x) $\rightarrow$ Q(x)) is vacuously True as long as there is at least one object in the universe for which P(x) is False (since F $\rightarrow$ T and F $\rightarrow$ F are True). The correct statement uses conjunction: $\exists x$ (Student(x) $\wedge$ Smart(x)).

Question 6: Given a binary predicate Likes(x, y), how do the semantic interpretations of $\forall x \exists y$ Likes(x, y) and $\exists y \forall x$ Likes(x, y) differ?

A) They are logically equivalent because quantifier ordering can always be swapped.

B) $\forall x \exists y$ Likes(x, y) means 'Everyone likes someone', whereas $\exists y \forall x$ Likes(x, y) means 'There is someone who is liked by everyone'.

C) $\forall x \exists y$ Likes(x, y) asserts that no two individuals can like the same person.

D) $\exists y \forall x$ Likes(x, y) asserts that everyone likes themselves.

Correct Answer: B) $\forall x \exists y$ Likes(x, y) means 'Everyone likes someone', whereas $\exists y \forall x$ Likes(x, y) means 'There is someone who is liked by everyone'.

Explanation: Passage [23] emphasizes that switching the order of universal and existential quantifiers alters the formula's meaning: $\forall x \exists y$ allows a different y for each x, whereas $\exists y \forall x$ requires one uniform y that satisfies all x.

Question 7: According to the quantifier negation rules (De Morgan's laws for quantifiers), which expression is logically equivalent to $\neg \forall x$ P(x)?

A) $\forall x \neg$ P(x)

B) $\exists x \neg$ P(x)

C) $\neg \exists x \neg$ P(x)

D) $\exists x$ P(x)

Correct Answer: B) $\exists x \neg$ P(x)

Explanation: Passages [18] & [19] define the quantifier negation identity: to negate a quantified formula, move the negation inside the scope and flip the quantifier ($\forall$ becomes $\exists$ and vice versa). Thus, $\neg \forall x$ P(x) $\equiv \exists x \neg$ P(x).

Question 8: In the FOL formula P(x) → ∃y Q(x, y), which variables are free and which are bound?

A) Both occurrences of x and y are bound.

B) y is bound by ∃y, while both occurrences of x are free.

C) x in P(x) is bound; x in Q(x, y) is free.

D) The formula contains no free variables and is therefore a ground sentence.

Correct Answer: B) y is bound by ∃y, while both occurrences of x are free.

Explanation: Passages [25] & [33] state that a variable is bound if it occurs within the scope of a quantifier introducing it. Here, y is bound by ∃y, whereas x is not governed by any quantifier, making both occurrences of x free variables.

Question 9: How is the statement 'At least 2 students attend the KR course' correctly formalized in FOL?

A) ∃x1 ∃x2 (Attend(x1, KR) ∧ Attend(x2, KR))

B) ∃x1 ∃x2 (Attend(x1, KR) ∧ Attend(x2, KR) ∧ x1 ≠ x2)

C) ∀x1 ∀x2 (Attend(x1, KR) → x1 = x2)

D) ∃! x1 Attend(x1, KR)

Correct Answer: B) ∃x1 ∃x2 (Attend(x1, KR) ∧ Attend(x2, KR) ∧ x1 ≠ x2)

Explanation: Passages [28] & [29] explain that simply writing ∃x1 ∃x2 (Attend(x1, KR) ∧ Attend(x2, KR)) allows x1 and x2 to refer to the exact same person. To guarantee at least 2 distinct individuals, explicit inequality x1 ≠ x2 must be included.

Question 10: When applying Existential Instantiation (Existential Elimination) to ∃x P(x), what constraint must be placed on the substituted constant symbol c?

A) c must be an arbitrary constant symbol already present in the Knowledge Base.

B) c must be a brand-new constant symbol (Skolem constant) that does not occur anywhere else in the KB.

C) c must be a universal variable bounded by ∀x.

D) c must be a zero-arity relation symbol.

Correct Answer: B) c must be a brand-new constant symbol (Skolem constant) that does not occur anywhere else in the KB.

Explanation: Passages [30] & [31] specify that Existential Instantiation requires replacing the variable with a brand-new constant symbol (a Skolem constant) not previously appearing in the KB to avoid asserting false identity with existing entities.

Question 11: What is the Most General Unifier (MGU) for the pair of atomic formulas P(a, x) and P(y, g(y))?

A) {a/y, g(a)/x} yielding common instance P(a, g(a))

B) {x/a, y/b} yielding common instance P(a, b)

C) FAIL due to occurs check error

D) {a/x, g(y)/y}

Correct Answer: A) {a/y, g(a)/x} yielding common instance P(a, g(a))

Explanation: Passage [35] provides the explicit unification table: unifying P(a, x) and P(y, g(y)) binds y ■ a and x ■ g(a), producing the MGU {a/y, g(a)/x} and most general common instance P(a, g(a)).

Question 12: In formal FOL semantics, how is an Interpretation I over a domain Δ defined for constants, n-ary functions, and n-ary predicates?

A) Constants map to {T, F}; functions map to relations; predicates map to variables.

B) Constants map to elements in Δ; n-ary functions map $\Delta^1 \rightarrow \Delta$; n-ary predicates map $\Delta^1 \rightarrow$ {T, F}.

C) Constants map to Δ^1; functions map to {T, F}; predicates map to integers.

D) Constants, functions, and predicates all map to continuous real vectors in Δ.

Correct Answer: B) Constants map to elements in Δ; n-ary functions map $\Delta^1 \rightarrow \Delta$; n-ary predicates map $\Delta^1 \rightarrow$ {T, F}.

Explanation: Passages [35] & [36] state that an interpretation assigns each constant to an object in domain Δ, each n-ary function to a mapping $\Delta^1 \rightarrow \Delta$, and each n-ary predicate to a mapping $\Delta^1 \rightarrow$ {T, F}.

Question 13: Which statement accurately describes the computational decidability of entailment in First-Order Logic?

A) General FOL entailment is fully decidable in polynomial time via truth tables.

B) General FOL entailment is semi-decidable; however, restricted fragments like Datalog (Horn without function symbols) and Description Logics are decidable.

C) General FOL entailment is completely undecidable under all subsets and fragments.

D) FOL entailment is decidable if sentences contain 3 or more variables, but undecidable with 2 variables.

Correct Answer: B) General FOL entailment is semi-decidable; however, restricted fragments like Datalog (Horn without function symbols) and Description Logics are decidable.

Explanation: Passage [39] explains that deciding satisfiability and entailment in general FOL is semi-decidable. Restricted subsets—such as Datalog (Horn clauses without function symbols) and Description Logics—are fully decidable.

Question 14: What fundamental expressive capability distinguishes Second-Order Logic from First-Order Logic?

A) FOL cannot represent disjunctions or material implications.

B) Quantifiers in FOL can only range over individual objects (terms), whereas Second-Order Logic allows quantifiers to range over predicates and functions.

C) FOL cannot use constant symbols or function terms.

D) Second-Order Logic is fully decidable, while FOL is undecidable.

Correct Answer: B) Quantifiers in FOL can only range over individual objects (terms), whereas Second-Order Logic allows quantifiers to range over predicates and functions.

Explanation: Passages [8] & [40] note that FOL restricts quantification to individual objects. Second-Order Logic extends expressiveness by allowing quantifiers to range over predicates and functions (e.g., $\forall x \forall y [(x = y) \leftrightarrow (\forall p p(x) \leftrightarrow p(y))]$).

Question 15: What does Gödel's Incompleteness Theorem (1931) establish regarding consistent formal logical systems capable of doing arithmetic?

A) Any consistent logic inference system must be incomplete.

B) Any complete logic inference system must be consistent.

C) Resolution refutation guarantees both consistency and completeness for Higher-Order Logic.

D) Consistency and completeness can be simultaneously achieved by adding finite inference rules.

Correct Answer: A) Any consistent logic inference system must be incomplete.

Explanation: Passages [42], [43], & [44] state Gödel's Incompleteness Theorem: 'Any consistent logic inference system MUST BE incomplete.' This proves a fundamental boundary on formal mathematical logic.